

Retail Customer Retention Analytics – Target

Power BI Project Report

Project Overview

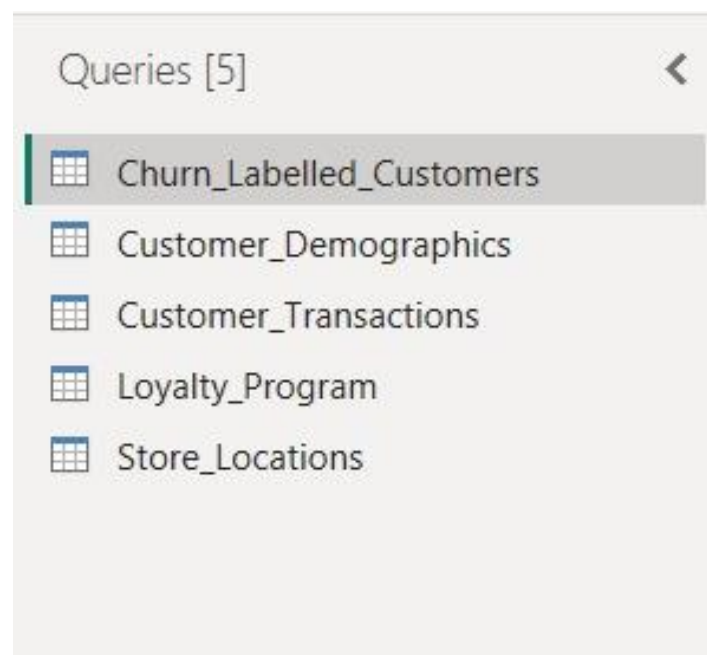
This project is based on Retail Customer Retention Analytics for Target using Power BI. The main purpose of this project is to understand customer purchasing behaviour and identify loyal, repeat, and churned customers. By analysing customer transactions, loyalty program details, store performance, and promotions, the dashboard provides useful business insights. Different charts, KPIs, and slicers are used to make the data easy to understand. This analysis helps identify areas where customer retention can be improved and supports better business decisions. It also provides a clear view of customer performance across different regions and sales channels.

Dataset Description

This project is completed using five different datasets. These datasets include customer details, transaction records, store information, loyalty program details, and churn data. All the datasets are imported and connected in Power BI to create a data model. After connecting the data, different reports and dashboards are created to analyse customer retention and understand customer behaviour more effectively.

Dataset names	Purpose
Customer_Demographics	Contains customer information such as age, gender, region, income level, membership date, and preferred shopping channel.
Customer_Transactions	Contains customer transaction details, purchase amount, product category, transaction date, and promotion information.
Store_Locations	Contains store information such as store type, region, and opening year.
Loyalty_Program	Contains loyalty tier, points earned, and points redeemed for each customer.
Churn_Labelled_Customers	Contains churn status, last purchase date, and churn reason for each customer. how can i make a box and write this all in it

Figure1:Datasets loaded into Power BI



Data Cleaning & Transformation

The datasets were imported into Power BI using Power Query. Basic data cleaning was performed to prepare the data for analysis. The data types were verified, and the required columns such as Membership_Duration, Transaction_Year, and Transaction_Month were created in Power Query. After the data was prepared, it was used to build the data model and create the Power BI dashboard.

Data Cleaning & Transformation Steps

- Imported all five CSV datasets into Power BI using Power Query.
- Verified the data types of all columns.
- Created the Membership_Duration column in Power Query.
- Extracted the Transaction_Year from the Transaction_Date column.
- Extracted the Transaction_Month from the Transaction_Date column.
- Prepared the datasets for building the data model.

Figure 2: Power Query Editor

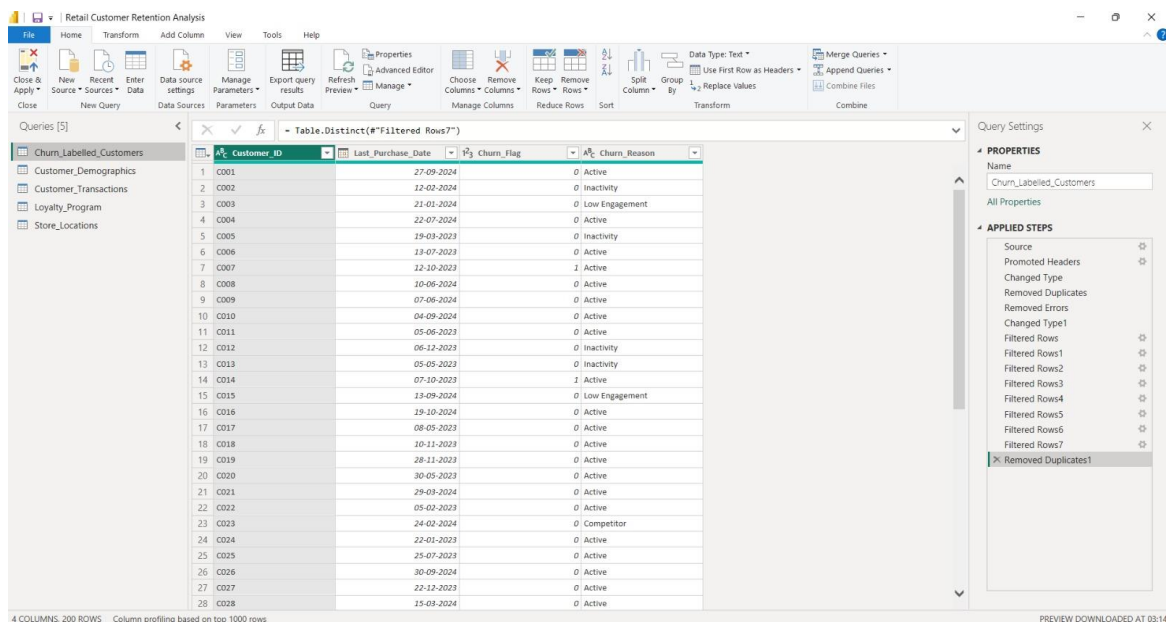


Figure 3: Applied Steps in Power Query

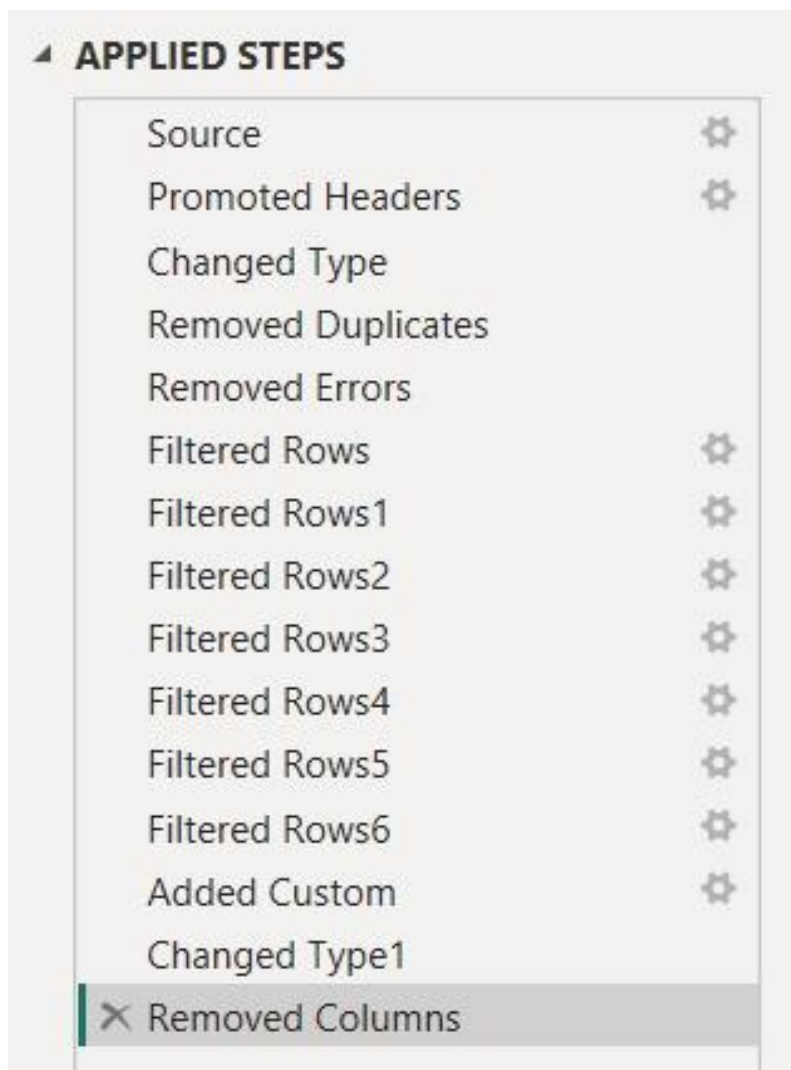
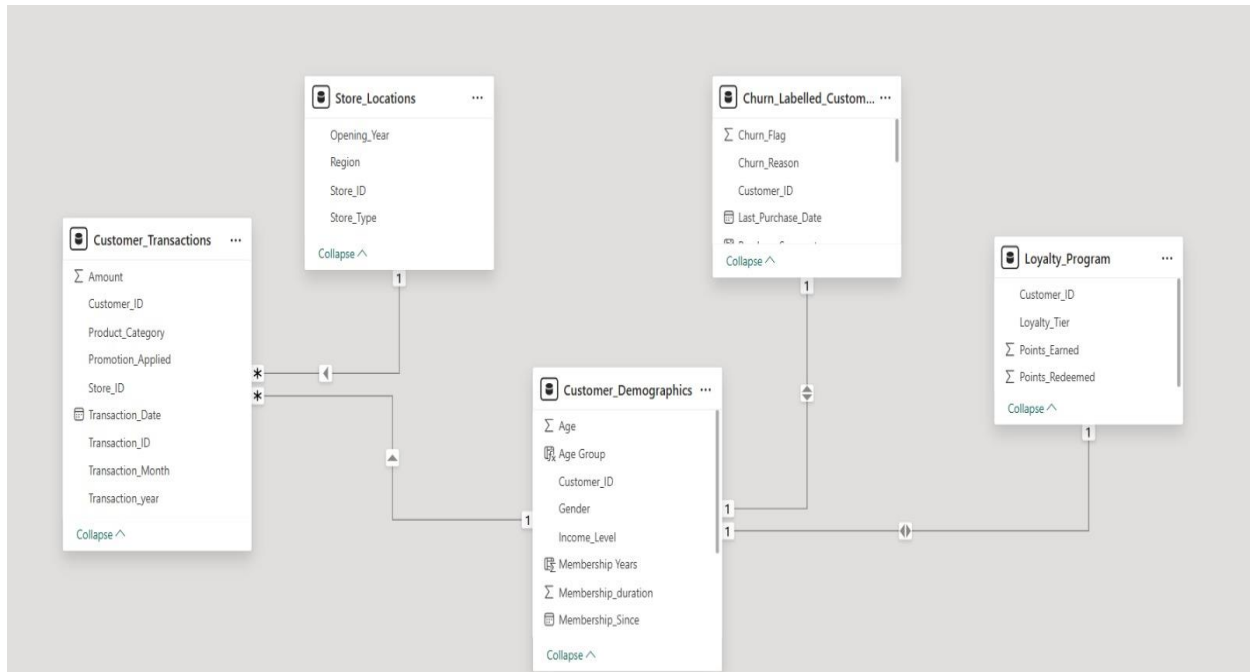


Figure 4: Created Columns in Power Query

Transaction_year	Transaction_Month	Membership_duration
2023	December	1050
2024	September	2481
2023	April	1215
2023	August	1829
2023	October	1390
2024	March	2803
2023	March	2467
2023	February	3050
2023	January	2380
2024	August	2580
2023	January	2632
2023	June	3078
2024	March	980
2024	April	2741
2023	May	2580
2023	October	1770
2023	October	2223
2023	July	2075
2023	April	2079
2024	February	1459
2024	July	2187
2023	September	695
2023	December	1331
2024	March	1574
2023	January	1199
2024	September	1890
2023	May	1451

Figure 5: Data Model in Power BI

A data model was created in Power BI by establishing relationships between the datasets using common fields such as Customer_ID and Store_ID. These relationships allow the data to be connected correctly and help create accurate reports and dashboards.



DAX Measures

DAX (Data Analysis Expressions) measures were created in Power BI to perform calculations and generate meaningful insights from the data. These measures help analyse customer retention, purchase behaviour, loyalty program performance, and overall business performance.

DAX Measures Create

Figure 6: Average Purchase Amount Measure

```
1 Avg Purchase Amount =  
2 AVERAGE('Customer_Transactions'[Amount])
```

Figure 7: Average Purchase Frequency Measure

```
1 Avg Purchase Frequency =  
2 AVERAGEX(  
3 VALUES('Customer_Demographics'[Customer_ID]),  
4 CALCULATE(COUNT('Customer_Transactions'[Transaction_ID]))  
5 )
```

Figure 8: Churn Rate Measure

```
1 Churn Rate =  
2 DIVIDE([Churned Customers],[Total Customers],0)
```

Figure 9: Churned Customers Measure

```
1 Churned Customers =  
2 CALCULATE(  
3 COUNTROWS(Churn_Labelled_Customers),  
4 Churn_Labelled_Customers[Churn_Flag] = 1  
5 )
```

Figure 10: Purchase Segment Measure

```
1 Purchase Segment =  
2 VAR Purchases =  
3 CALCULATE(  
4 COUNT(Customer_Transactions[Transaction_ID])  
5 )  
6 RETURN  
7 SWITCH(  
8 TRUE(),  
9 Purchases <= 3, "Low-Tier",  
10 Purchases <= 8, "Mid-Tier",  
11 "High-Tier"  
12 )
```

Figure 11: Repeat Customer Measure

```
1 Repeat Customers =  
2 COUNTROWS(  
3 FILTER(  
4 VALUES(Customer_Transactions[Customer_ID]),  
5 CALCULATE(COUNT(Customer_Transactions[Transaction_ID])) > 1  
6 )  
7 )
```

Figure 12: Total Customer measure

```
1 Total Customer = COUNT(Customer_Demographics[Customer_ID])
```

Figure 13: CLV Measure

```
1 CLV =  
2 DIVIDE(  
3 SUM('Customer_Transactions'[Amount]),  
4 SUM('Customer_Demographics'[Membership_Years])  
5 )
```

Figure 14: Promotion Transaction Measure

```
1 Promotion Transaction % =  
2 DIVIDE(  
3 COUNTROWS(  
4 FILTER(  
5 'Customer_Transactions',  
6 'Customer_Transactions'[Promotion_Applied] = "Yes"  
7 )  
8 ),  
9 COUNTROWS('Customer_Transactions')  
10 )
```

Figure 16: Purchase Tier Measure

```
1 Purchase Tier =  
2 SWITCH(  
3 TRUE(),  
4 [Purchase Count] <= 3, "Low-Tier",  
5 [Purchase Count] <= 8, "Mid-Tier",  
6 "High-Tier"  
7 )
```

Figure 17: Purchase Count

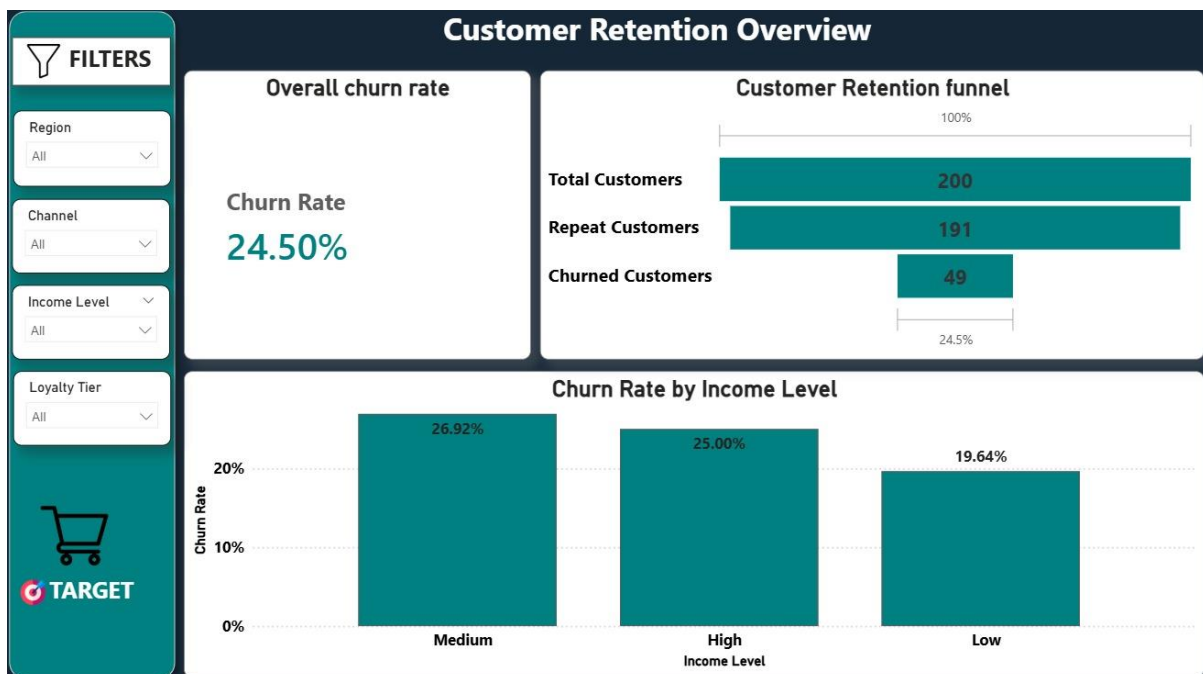
```
1 Purchase Count =  
2 COUNTROWS(  
3 RELATEDTABLE('Customer_Transactions')  
4 )
```

Dashboard Analysis

This project includes four dashboards created in Power BI to present the analysis in a simple and interactive way. Each dashboard focuses on a different part of customer retention, such as customer behaviour, loyalty program performance, store performance, and business insights. Different charts, KPIs, and slicers are used to make the information easy to understand and analyse.

Dashboard 1: Customer Retention Overview

This dashboard gives an overview of customer retention. It shows the overall churn rate, the number of total, repeat, and churned customers, and the churn rate based on different income levels. Filters such as region, channel, income level, and loyalty tier are included to make the analysis more interactive.

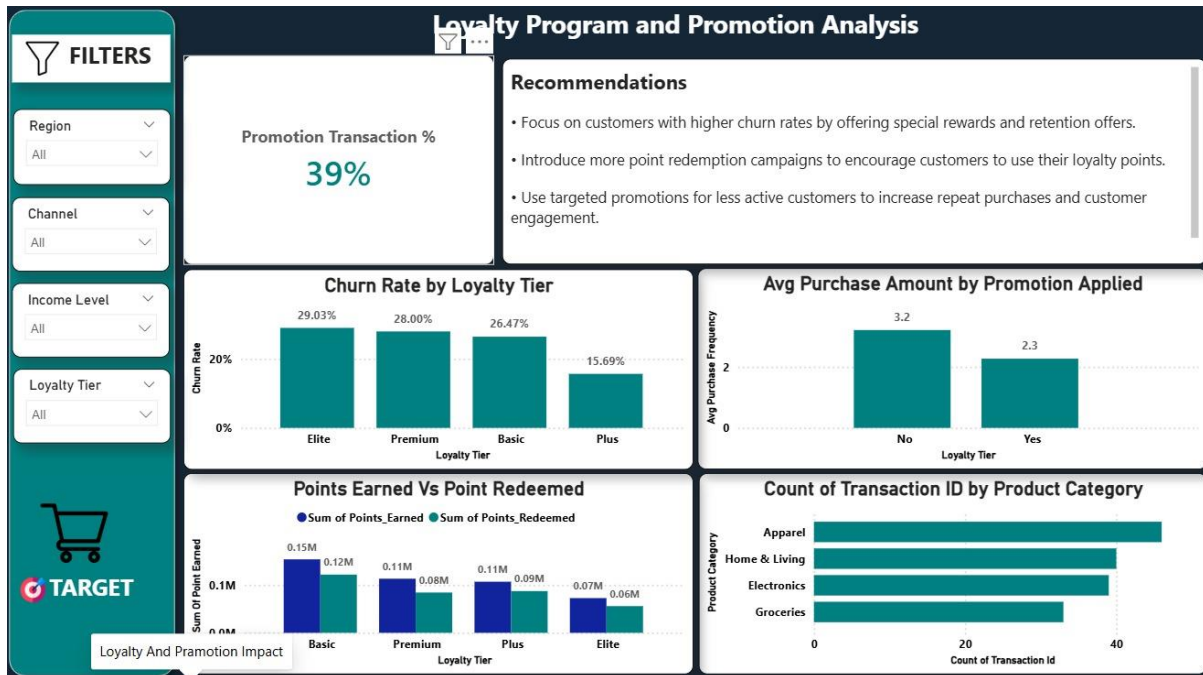


Key Insights

- The overall churn rate is 24.50%.
- Most customers are repeat customers.
- Churn is higher for customers with a Medium income level.
- Customers with a Low income level have the lowest churn.
- The filters help compare customer data in different ways.

Dashboard 2: Loyalty Program and Promotion Analysis

This dashboard focuses on the loyalty program and promotional activities. It shows the promotion transaction percentage, churn rate for different loyalty tiers, points earned and redeemed by customers, average purchase amount based on promotions, and transaction count by product category. The filters help users analyze the data for different customer groups.

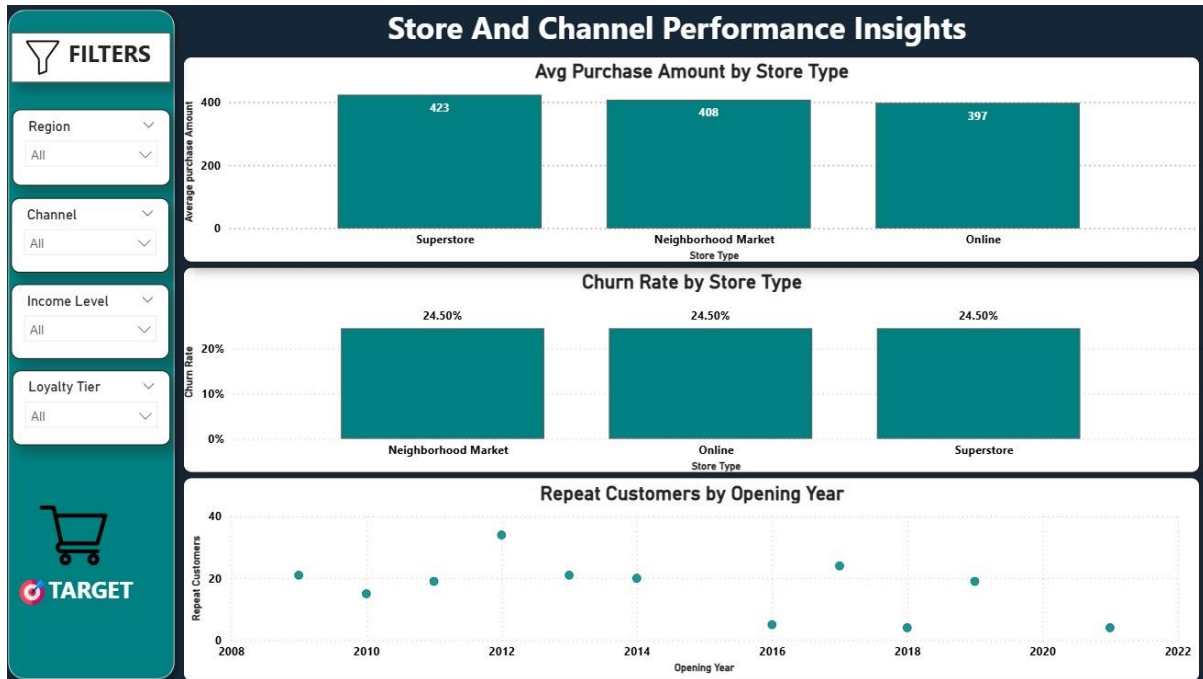


Key Insights

- 39% of the transactions were made using promotions.
- The Elite loyalty tier has the highest churn rate, while the Plus tier has the lowest.
- Customers have earned more loyalty points than they have redeemed.
- Apparel has the highest number of transactions among all product categories.
- The dashboard helps compare loyalty program performance and promotions using different filters.

Dashboard 3: Store and Channel Performance Insights

This dashboard shows the performance of different store types and customer channels. It includes the average purchase amount, churn rate by store type, and repeat customers based on the store opening year. The filters on the left help users compare the data for different customer groups.

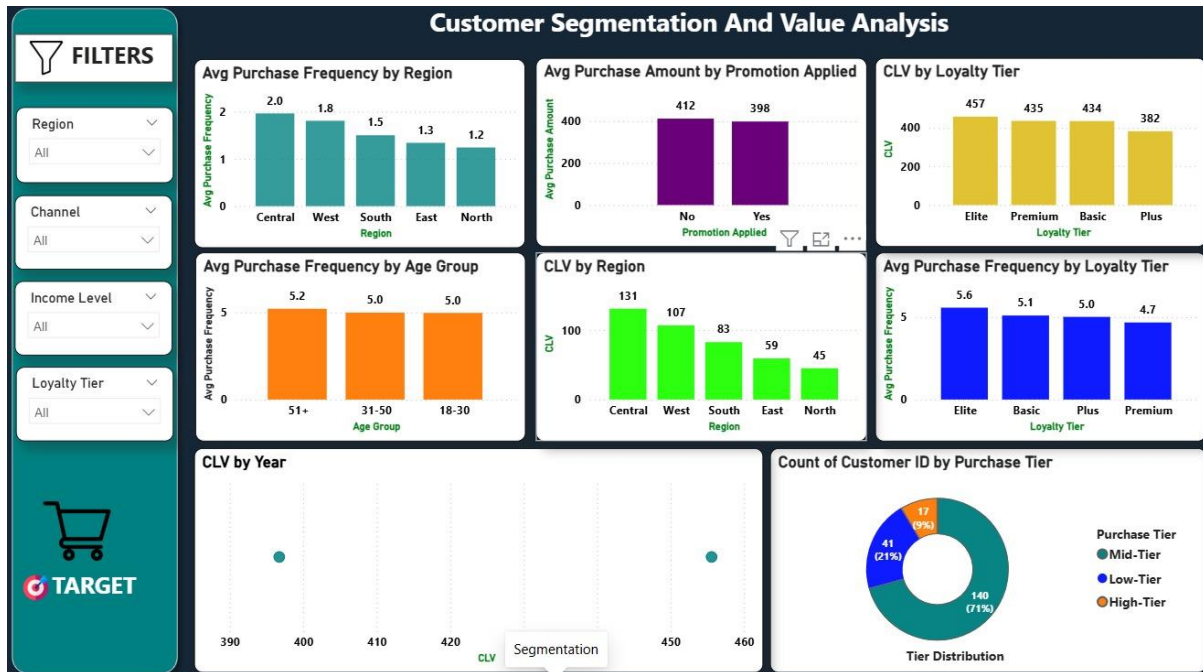


Key Insights

- Superstores have the highest average purchase amount (423).
- Online stores have the lowest average purchase amount (397).
- The churn rate is 24.50% across all store types.
- The number of repeat customers varies depending on the store opening year.
- The filters make it easy to compare store performance for different customer groups.

Dashboard 4: Customer Segmentation and Value Analysis

This dashboard shows customer segmentation based on different factors such as region, age group, loyalty tier, and purchase tier. It also includes customer lifetime value (CLV), average purchase frequency, and average purchase amount. The dashboard helps compare customer behaviour and identify valuable customer segments.



Key Insights

- The Central region has the highest customer lifetime value (CLV).
- Customers in the 51+ age group have the highest average purchase frequency.
- The Elite loyalty tier has the highest CLV.
- Most customers belong to the Mid-Tier purchase segment.
- The dashboard helps compare customer value across different regions and loyalty tiers.

Executive Summary

This project was completed using Power BI to understand customer retention at Target. Five datasets were used in this project, including customer details, transaction records, loyalty program data, store information, and churn data. The data was cleaned in Power Query, and DAX measures were created to perform the required calculations. Finally, interactive dashboards were developed to analyse customer behaviour and present the results in a simple and easy-to-understand way.

Recommendations

- More attention can be given to customers who have stopped purchasing to encourage them to shop again.
- The loyalty program can be made more attractive so that more customers use their reward points.
- Stores and regions with lower customer activity can be reviewed to understand the reasons and improve performance.
- Promotional offers can be planned based on customer preferences and buying patterns.
- The Power BI dashboard can be updated regularly to track customer trends and support future business decisions.

Conclusion

This project helped me understand how to use Power BI for data analysis. I learned how to clean data, create a data model, write DAX measures, and build interactive dashboards. The project also helped me understand customer retention and how data can be used to support business decisions.

Project Demonstration Video

Google Drive Link: https://drive.google.com/file/d/1-a9shlea7Lgt4NpCHDCRdty5_M7rlCUK/view?usp=sharing